



PVDepth: Panoramic Video Depth Estimation via Geometry-aware Spatiotemporal Adaptation

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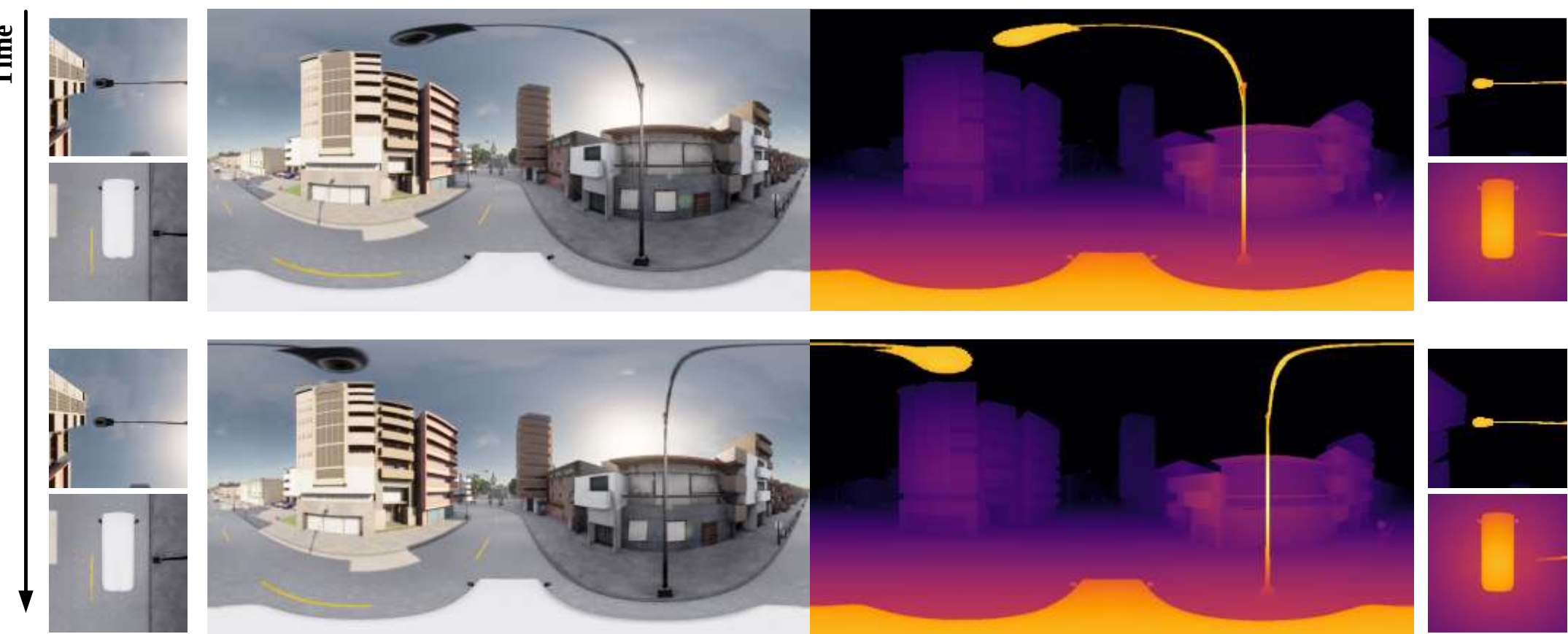


TL;DR: End-to-end panoramic video depth estimation with a large-scale RGB-D dataset and geometry-aware spatiotemporal adaptation.

Challenges in Panoramic Video Depth Estimation

Data scarcity: there is no public large-scale panoramic RGB-D video benchmark.

Panoramic-perspective domain gap:
ERP-specific spatial distortion
Non-linear temporal dynamics



ERP stretches polar regions and induces irregular temporal dynamics.

Why Existing Methods Fall Short

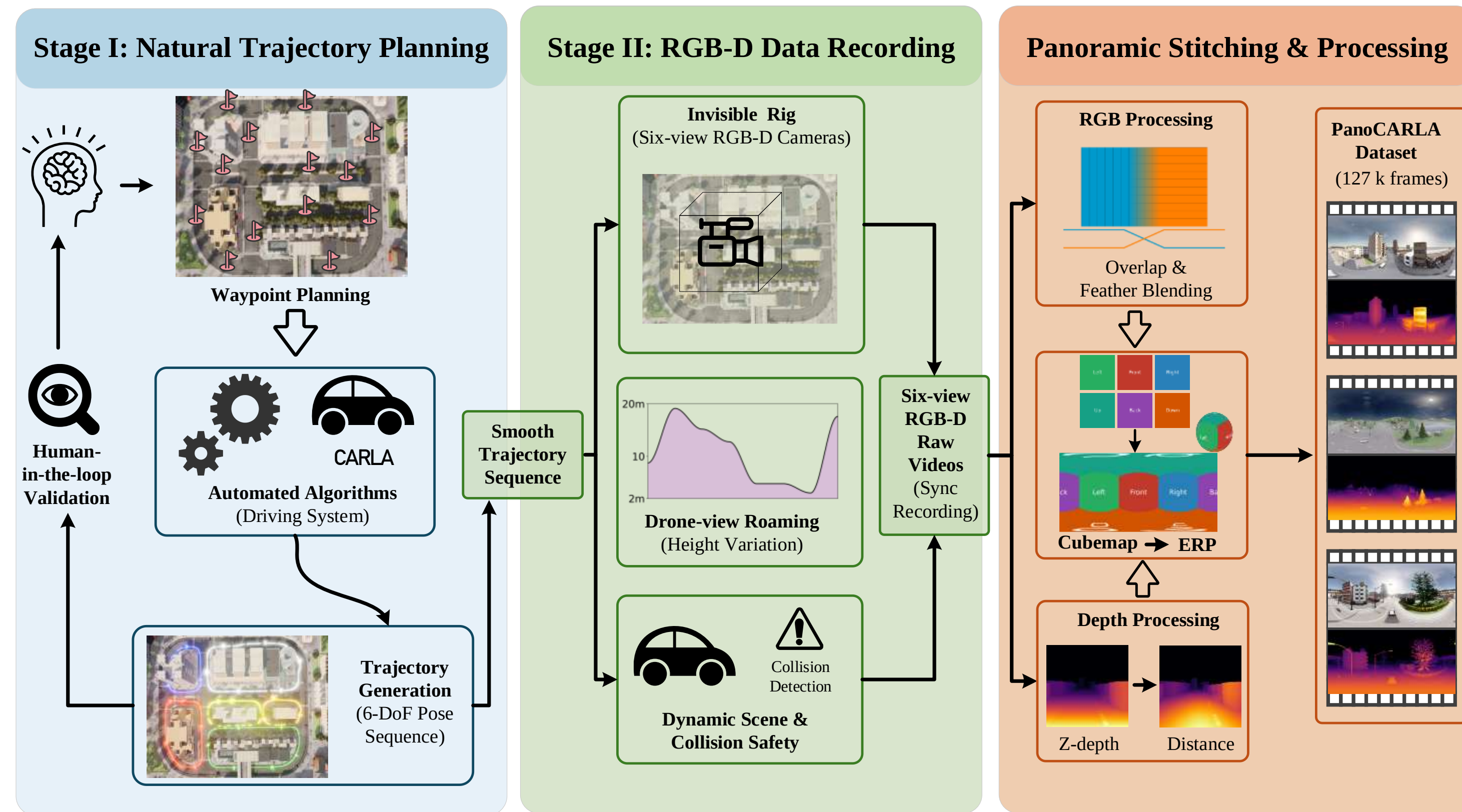
Panoramic image-based methods:
good per-frame results, bad temporal consistency.

Post-processing alignment methods:
computationally expensive and struggles with dynamic scenes.

Perspective video depth methods:
sub-optimal adaptation caused by spatiotemporal domain gap.

Need: training data && panoramic geometry-aware adaptation

PanoCARLA Dataset



- ✦ Built in the CARLA simulator.
- ✦ RGB, depth, and camera poses.
- ✦ Outdoor scenes with dynamic objects.
- ✦ 50.61 km drone-view roaming.

Dataset	Structure -3D	Matterport -3D	Stanford -2D3D	Omni360 -Scene	PanoCARLA
Is Video?	×	×	×	×	✓
#Frames	196K	10K	1K	61K	126.7K

Data scarcity



PanoCARLA

Spatial distortion



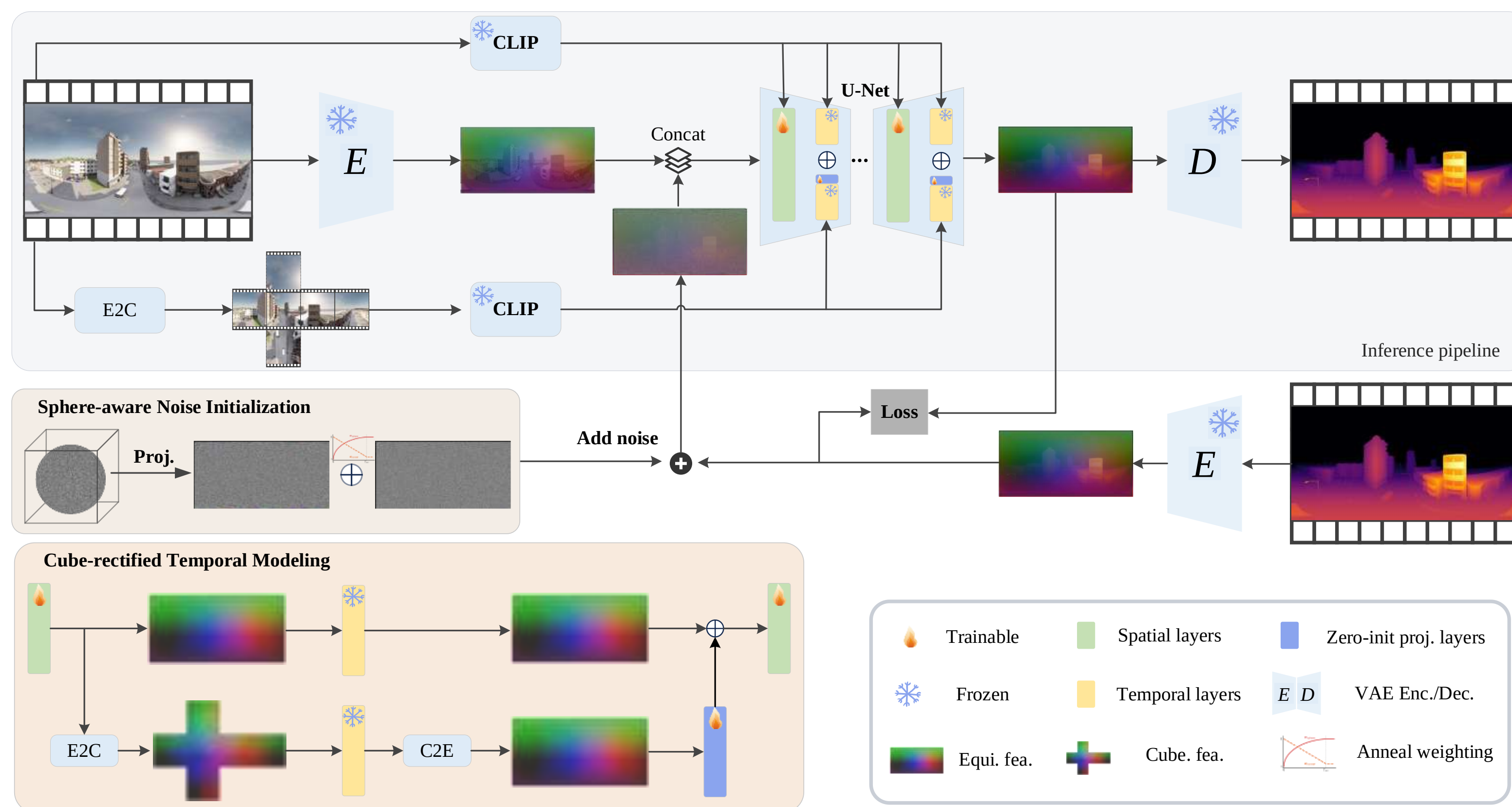
PSNI

Temporal nonlinearity



CRTM

PVDepth Framework



Built upon DepthCrafter.

PSNI (Progressive Sphere-aware Noise Initialization strategy):
Anneal noise from planar to spherical.
Adapt to non-uniform ERP density.

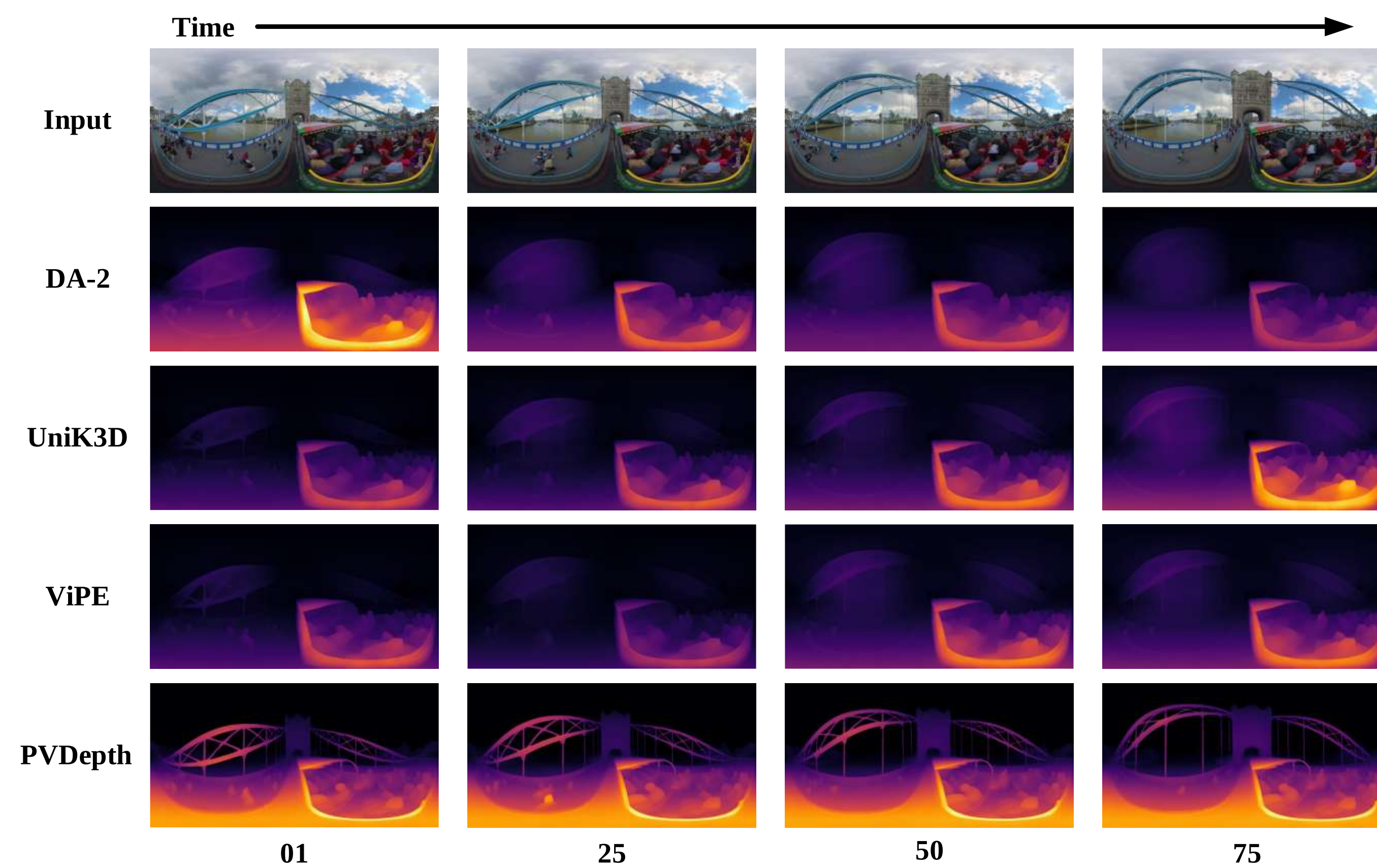
CRTM (Cube-rectified Temporal Modeling module):
An auxiliary cubemap temporal branch.
Rectify non-linear temporal dynamics.

Experiments

Quantitative comparison (avg over all settings)

Method	AbsRel ↓	δ_1 ↑
Panorama image depth-based		
DA-2 _{ZS} (Li et al., 2025)	1.171 (0.757)	0.615 (0.754)
UniK3D _{ZS} (Piccinelli et al., 2025)	0.592 (0.461)	0.638 (0.736)
PanoDA _{ZS} (Cao et al., 2025)	0.710 (0.526)	0.280 (0.342)
DepthAnywhere _{ZS} (Wang & Liu, 2024)	0.449 (0.233)	0.464 (0.800)
DepthAnywhere _{FT} (Wang & Liu, 2024)	0.458 (0.207)	0.483 (0.846)
Post-processing alignment-based		
Matrix-3D (Yang et al., 2025b)	0.494	0.513
ViPE (UniK3D-based) (Huang et al., 2025)	0.670	0.721
Perspective video depth-based		
VideoDepthAnything _{ZS} (Chen et al., 2025)	0.356	0.505
DepthAnyVideo _{ZS} (Yang et al., 2025a)	0.295	0.608
DepthCrafter(Cube-based) _{ZS}	0.941	0.328
DepthCrafter(Cube-based) _{FT}	0.417	0.590
DepthCrafter _{ZS} (Hu et al., 2025)	0.290	0.567
DepthCrafter _{FT} (Hu et al., 2025)	<u>0.232</u>	<u>0.754</u>
PVDepth (Ours)	0.199	0.798

Qualitative comparison on In-the-Wild Videos



PVDepth achieves higher accuracy and temporal stability.

Ablation study (avg over all settings)

Method	PSNI	CRTM	AbsRel ↓	δ_1 ↑
Baseline	—	—	0.248	0.757
w/ PSNI	✓	—	0.228	0.782
w/ CRTM	—	✓	0.208	0.779
PVDepth (Full)	✓	✓	0.199	0.798